Fiscal Year:	FY 2015	**Task Last Updated:**	FY 08/01/2019
PI Name:	Alberts, Jeffrey Ph.D.		
Project Title:	Individual and Group Behavior of Bion-M1 Rodents		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Bloomington	**State:**	IN
Zip Code:	47405-7007	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	06/30/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Klotz, Rebecca	**Contact Phone:**	650-604-1119
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Flight Program:	Bion		
Flight Assignment:	Bion M1 NOTE: Extension was to 6/30/2015 per NSSC information (Ed., 5/5/15) NOTE: Extended to 7/31/2015 per A. Chu/ARC; original end date was 7/31/2013 (Ed., 7/14/2014) NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2014)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AP12G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The in-flight video recording capabilities planned for the Bion-M1 Project make possible a new generation of science on unmanned satellites. The present proposal is to analyze rodent behavior across the one-month duration of the orbital flight of Bion-M1. The specific, testable hypotheses for Bion-M1 include microgravity effects on the kinds and amounts of behavior that rodents will express. General activity and its circadian pattern will be examined, and specific functional categories of individual and group social behaviors will be characterized and quantified. Kinematic measures of limb and trunk movements will also be taken. Modification of individual and group behavior can alter immune, endocrine, and physiological function. By integrating behavioral data with those from the same animals studied by various primary and secondary study PIs, it will be possible to test alternative explanations for alterations in immune, musculoskeletal, neural, and endocrine changes with spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The planned scientific studies for Bion M-1 are primarily in the realm of basic, "fundamental biology," but the studies will be pioneering in many ways and are capable of spawning exciting Earth benefits, mostly in the realm of insights into fundamental biological processes related to immune function, muscle and bone integrity, and adaptation of the nervous system.
Task Progress:	[Ed. note: compiled in July 2019 from PI final report submitted to the NASA Shared Services Center in November 2015 ; received in July 2019 by NASA Ames and then to Task Book] The Bion M-1 mission presented an exciting opportunity for studying adult mice during a relatively long-duration (one month) exposure to spaceflight conditions and immediately upon recovery on Earth. The design of this Russian flight experiment was based on group-housed male mice (three adult male, C57/Bl6 mice per compartment). Some of the mice were equipped with implanted telemetry equipment that provided heart rate and blood pressure data. Significantly for the present report, the Bion compartments provided on-board video surveillance of the mice, with significant volume of daily storage. Upon landing, the present effort included post-flight behavioral testing of the mice. Conventional wisdom has long held that unrelated adult male mice cannot be housed together without competition over resources such as space and access to resources, pathological stress, and fighting. NASA has incorporated such concerns into their life sciences operations, thus limiting the use of this valuable mammalian model for flight-based investigations. Bion M-1 procedures and results contributed an important alteration to the conventional wisdom, and the video data and direct observations that comprised the present investigations provide valuable information. We learned that the Russian scientists employed several innovative, yet simple techniques to produce triads of unrelated adult male mice that lived amicably in the small housing compartments for one month. This was accomplished by establishing, in their laboratory vivarium, triads of adult males. They were housed in a standard cage and provided a shelter apparatus to share. Mice that did not share the shelter were thus screen and eliminated from the pool. There followed a regime of daily handling. These animals were also maintained at about 90% ad lib body weight for purposes of operant training. These procedures yielded groups of adult males that lived together amicably in the Bion habitats in microgravity and under Earth-based control conditions. Their excellent performance violated the conventional wisdom and provides an important example of how we might well use group-housed male mice in future studies.
Bibliography Type:	Description: (Last Updated: 08/01/2019)
Articles in Peer-reviewed Journals	Andreev-Andrievskiy A, Popova A, Boyle R, Alberts J, Shenkman B, Vinogradova O, Dolgov O, Anokhin K, Tsvirkun D, Soldatov P, Nemirovskaya T, Ilyin E, Sychev V. "Mice in Bion-M 1 space mission: Training and selection." PLoS One. 2014 Aug 18;9(8):e104830. https://doi.org/10.1371/journal.pone.0104830 ; PubMed PMID: 25133741; PubMed Central PMCID: PMC4136787 , Aug-2014